AlZheimer disease

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THE PROBLEM

Alzheimer’s is hard to detect early because its symptoms develop slowly and are often unnoticed.

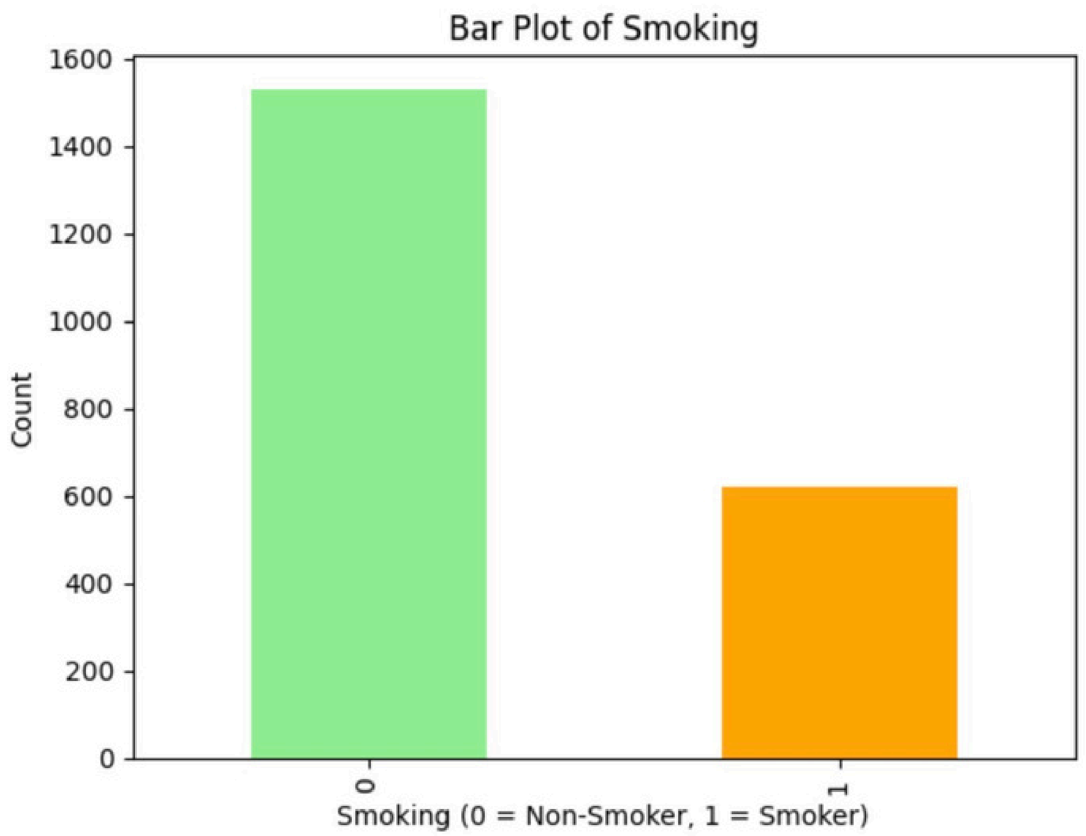
WHY ITS IMPORTANT

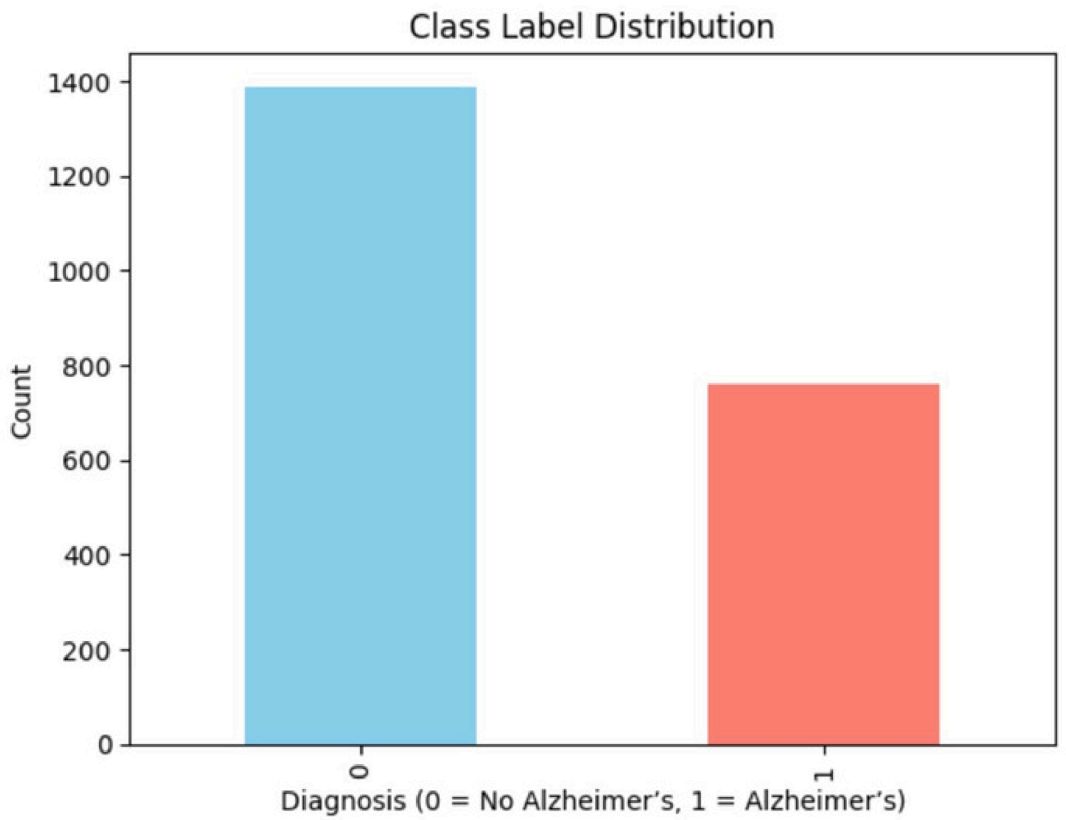
Early detection improves treatment planning and makes diagnosis more reliable.

DATA OVERVIEW

- Source: Kaggle Alzheimer's Dataset
- Number of features : 35
- Number of records : 2149
- Class label: Diagnosis (0 = No Alzheimer's, 1 = Alzheimer's)
- Other features: include age, gender, BMI, blood pressure, cholesterol,
sleep quality, physical activity, and family history of Alzheimer'

DATA REPRESENTION





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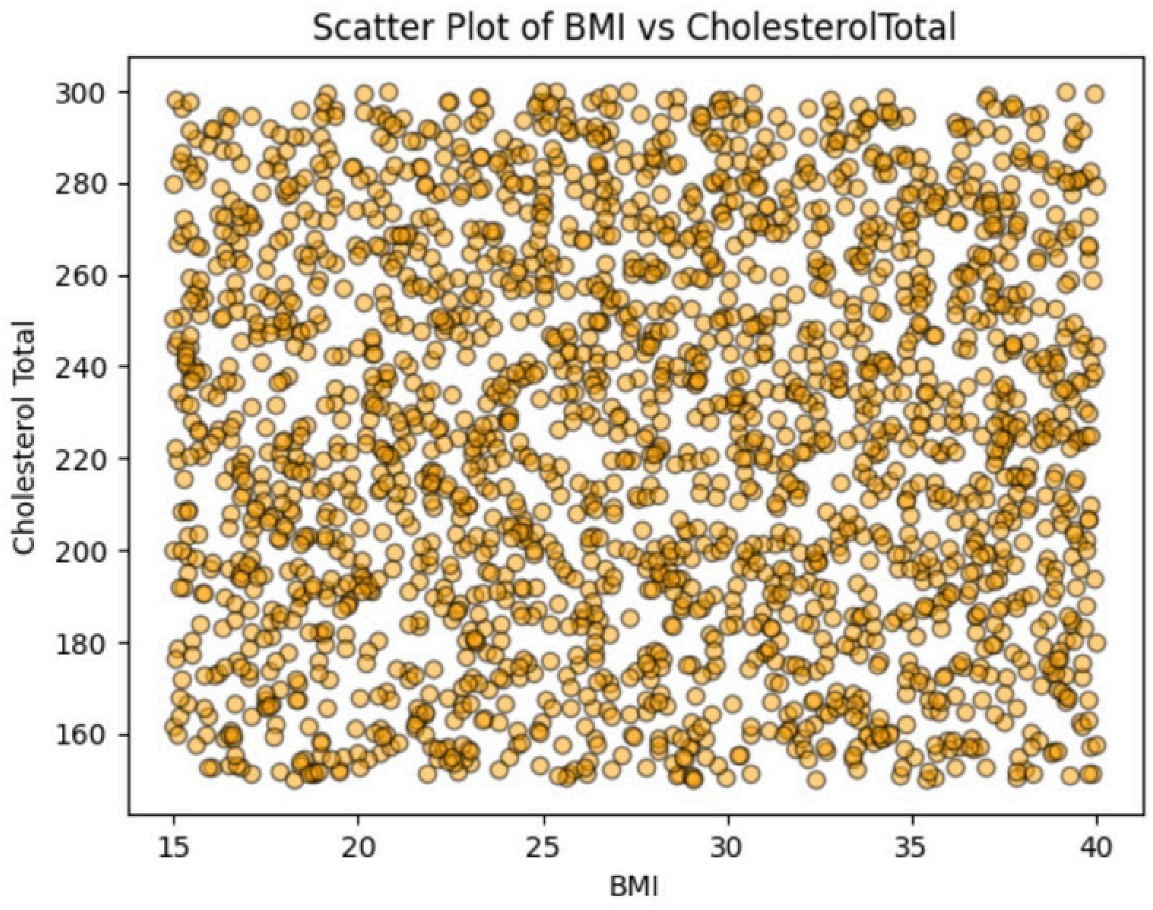

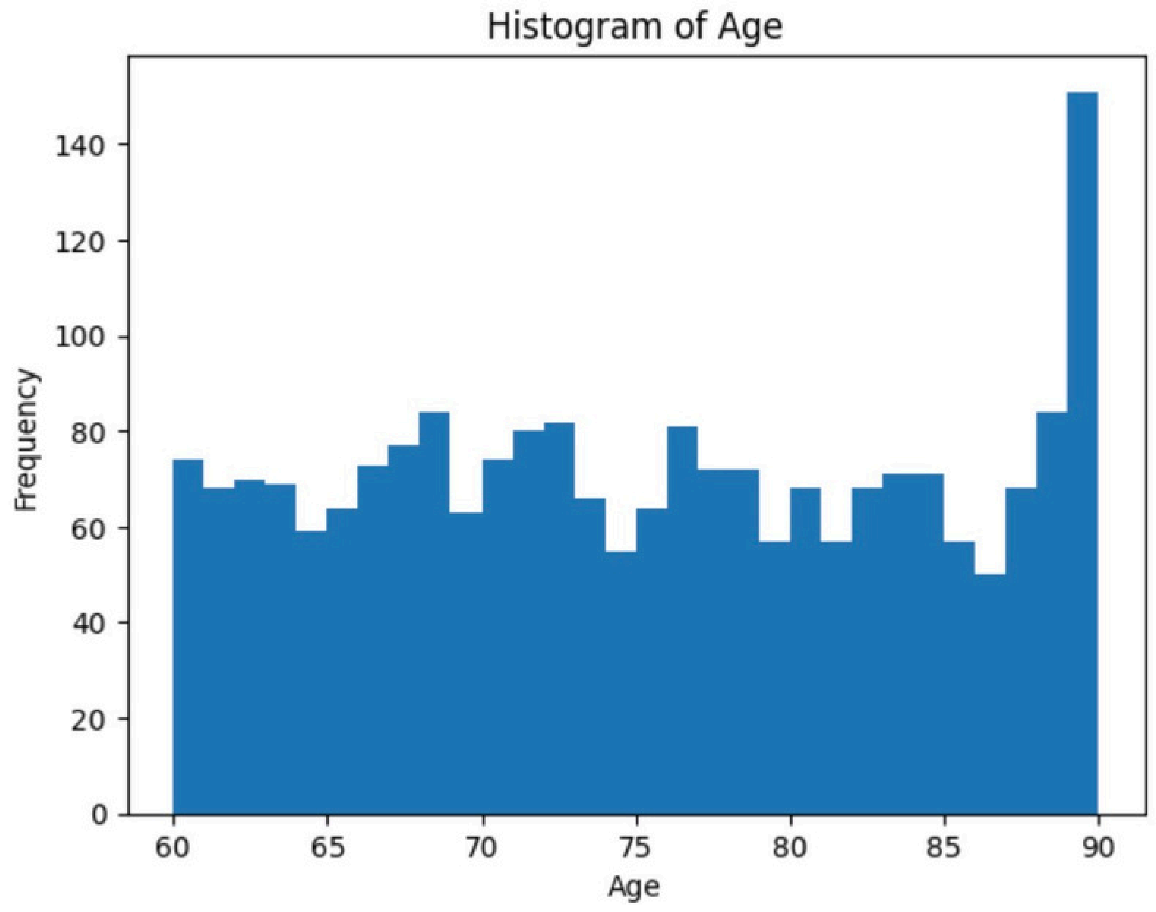

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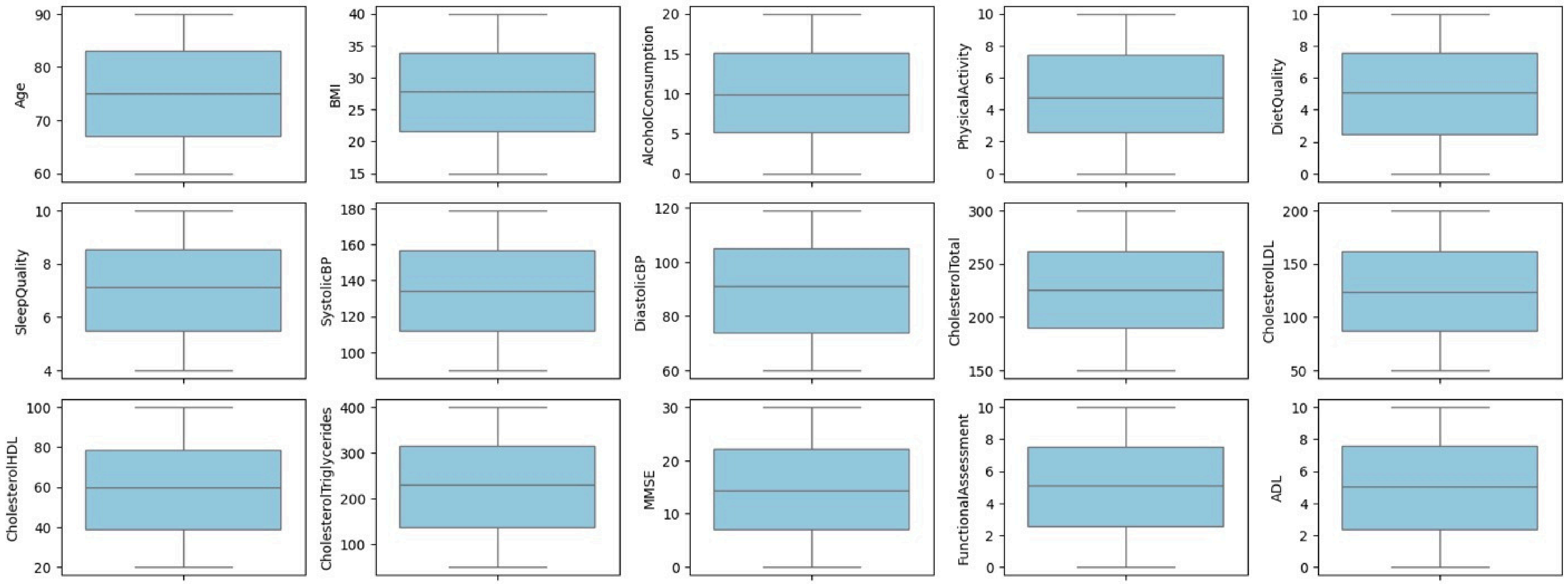


DATA PREPROCESSING

Data cleaning

- No cleaning was needed the data is already clean

Data Transmission

- Normalization
- Discretization

Data Reduction

- Feature selection

DISCRETIZATION

	Age	AgeGroup
1033	88	80+
634	89	80+
401	85	80+
305	89	80+
1861	71	70–79

converted continuous Age into an AgeGroup feature using fixed ranges (<60, 60–69, 70–79, 80+)

NORMALIZTION

	BMI	AlcoholConsumption	PhysicalActivity	DietQuality	SleepQuality	SystolicBP	DiastolicBP	CholesterolTotal	CholesterolTotal	CholesterolLDL	CholesterolHDL	CholesterolTriglycerides	MMSE	FunctionalAssessment	ADL
674	0.970115	0.672582	0.416137	0.035296	0.607735	0.191011	0.915254	0.392081	0.392081	0.520918	0.949786	0.222307	0.397491	0.010886	0.813904
194	0.854952	0.712238	0.996042	0.358057	0.253107	0.662921	0.084746	0.856150	0.856150	0.308395	0.704705	0.625823	0.790476	0.179845	0.076228
1766	0.585430	0.565385	0.814911	0.998192	0.687419	0.887640	0.728814	0.924254	0.924254	0.125815	0.661226	0.188979	0.941062	0.370923	0.005078
2067	0.628048	0.427371	0.296981	0.315915	0.065677	0.235955	0.779661	0.495605	0.495605	0.078628	0.296139	0.731665	0.133470	0.464217	0.715179
1100	0.160013	0.723333	0.508038	0.982745	0.670107	0.483146	0.084746	0.056488	0.056488	0.907957	0.642283	0.925013	0.579686	0.026388	0.317566

We applied Min–Max normalization to continuous numerical features only

FEATURE SELECTION

	PatientID	Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury	Hypertension	...	CholesterolHDL	CholesterolTriglycerides	MMSE	FunctionalAssessment	MemoryComplaints	BehavioralProblems	ADL	Disorientation	PersonalityChanges	Diagnos
1577	6328	0	2	0.647154	0.242093	0	0	0	0	0	...	0.891492	0.458576	0.429205	0.742090	0	0	0.169977	0	0	
1677	6428	0	1	0.889123	0.531558	1	0	0	0	0	...	0.302137	0.947092	0.219524	0.453590	1	0	0.811778	0	0	
1539	6290	0	1	0.370524	0.356954	0	0	0	0	0	...	0.009865	0.013415	0.655996	0.588209	0	0	0.020655	0	1	
62	4813	1	3	0.452582	0.011974	1	1	0	0	0	...	0.200053	0.433209	0.020473	0.759066	0	1	0.782637	0	0	
1548	6299	0	1	0.523948	0.474436	0	0	0	0	0	...	0.769914	0.579249	0.284214	0.432761	0	0	0.413540	0	0	

Categorical features were encoded using one-hot encoding, and the target Diagnosis was converted to numeric codes

BEFORE PREPROCESSING

	PatientID	Age	Gender	Ethnicity	EducationLevel	BMI	Smoking	AlcoholConsumption	PhysicalActivity	DietQuality	...	MemoryComplaints	BehavioralProblems	ADL	Confusion	Disorientation	PersonalityChanges	DifficultyCompletingTasks	Forgetfulness	Diagnosis	DoctorInCharge
0	4751	73	0	0	2	22.927749	0	13.297218	6.327112	1.347214	...	0	0	1.725883	0	0	0	1	0	0	XXXConfid
1	4752	89	0	0	0	26.827681	0	4.542524	7.619885	0.518767	...	0	0	2.592424	0	0	0	0	1	0	XXXConfid
2	4753	73	0	3	1	17.795882	0	19.555085	7.844988	1.826335	...	0	0	7.119548	0	1	0	1	0	0	XXXConfid
3	4754	74	1	0	1	33.800817	1	12.209266	8.428001	7.435604	...	0	1	6.481226	0	0	0	0	0	0	XXXConfid
4	4755	89	0	0	0	20.716974	0	18.454356	6.310461	0.795498	...	0	0	0.014691	0	0	1	1	0	0	XXXConfid

AFTER PREPROCESSING

	PatientID	Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury	Hypertension	...	CholesterolHDL	CholesterolTriglycerides	MMSE	FunctionalAssessment	MemoryComplaints	BehavioralProblems	ADL	Disorientation	PersonalityChanges	Diagnosis
0	4751	0	2	0.316960	0.837564	0	0	1	0	0	...	0.171039	0.319802	0.715606	0.652102	0	0	0.172486	0	0	0
1	4752	0	0	0.473058	0.525021	0	0	0	0	0	...	0.738026	0.698711	0.687251	0.712108	0	0	0.259154	0	0	0
2	4753	0	1	0.111553	0.945597	1	0	0	0	0	...	0.622290	0.095072	0.245145	0.589697	0	0	0.711936	1	0	0
3	4754	1	1	0.752163	0.731994	0	0	0	0	0	...	0.605851	0.649922	0.466410	0.896823	0	1	0.648094	0	0	0
4	4755	0	0	0.228472	0.265892	0	0	0	0	0	...	0.461019	0.688892	0.450619	0.604699	0	0	0.001341	0	1	0

DATA MINING TECHNIQUES

CLASSIFICATION

- **Information Gain (Entropy)**
- **GINI INDEX**

CLUSTERING

- **SILHOUETTE SCORE**
- **INERTIA** (AKA within-cluster sum of squares)
- **K-means**

DATA MINING TECHNIQUES

• CLASSIFICATION

To evaluate the model's performance under different data conditions, we divided the dataset into three train–test splits:

- 90% training with 10% testing
- 80% training with 20% testing
- 70% training with 30% testing

• CLUSTERING

To determine an appropriate number of clusters for our dataset, we evaluated three different values of **K: 2, 3, and 4.**

EVALUATION & COMPARISON

CLASSIFICATION RESULTS

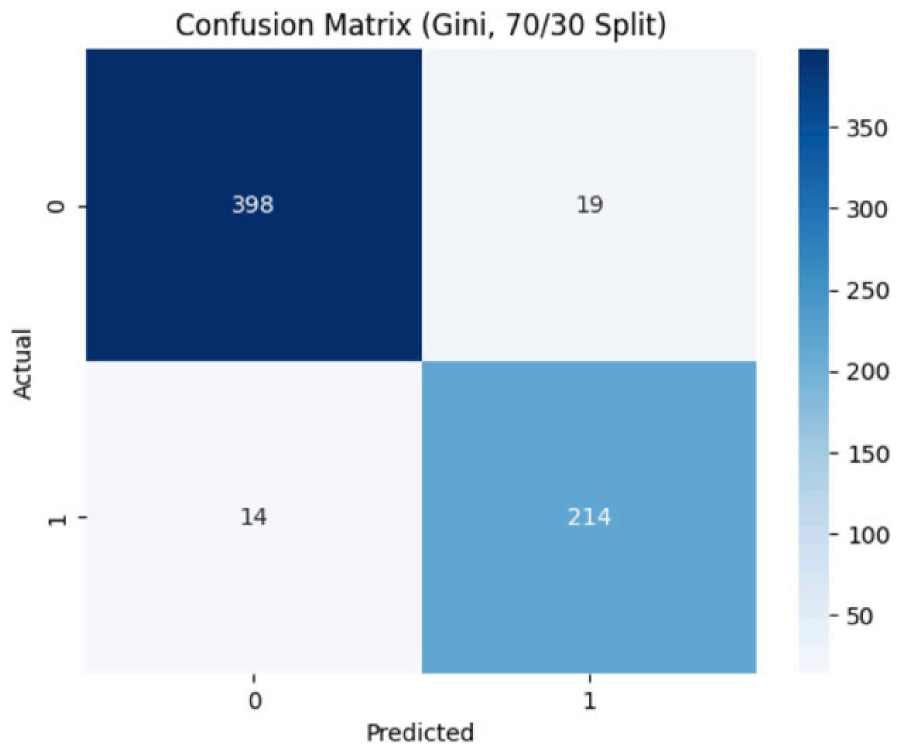
The accuracy results for each configuration are summarized below

Split Ratio	Criterion	Accuracy
90/10	Gini	0.9488
90/10	Entropy	0.9488
80/20	Gini	0.9465
80/20	Entropy	0.9348
70/30	Gini	0.9488
70/30	Entropy	0.9410

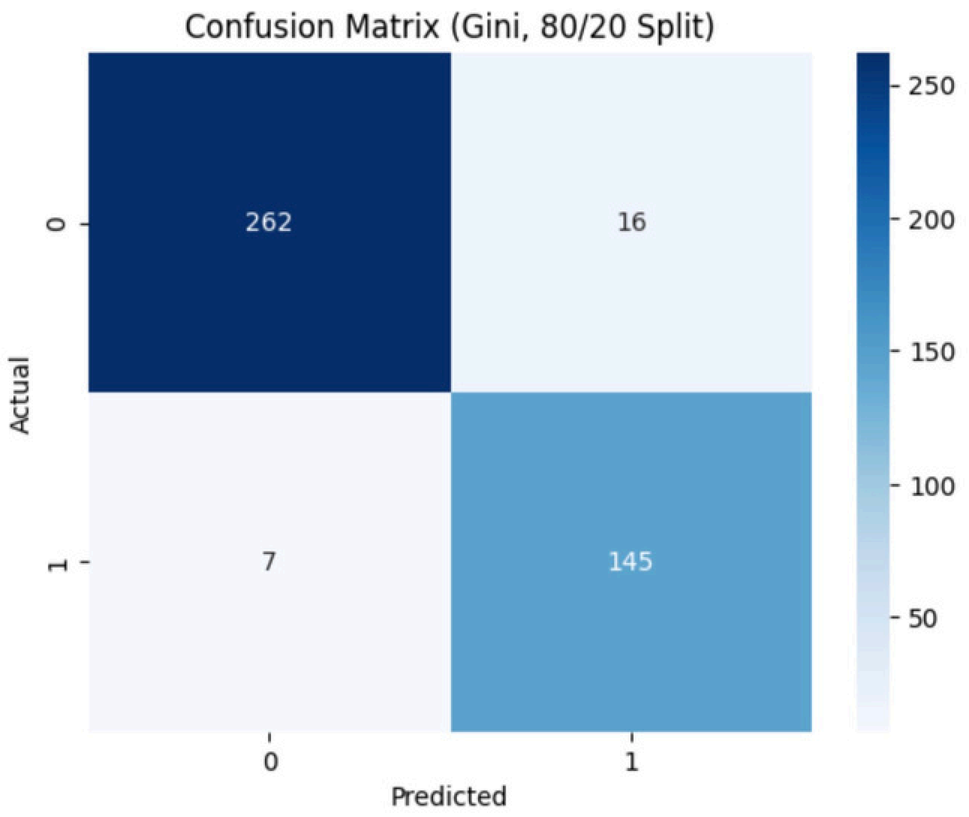
EVALUATION & COMPARISON

CLASSIFICATION RESULTS

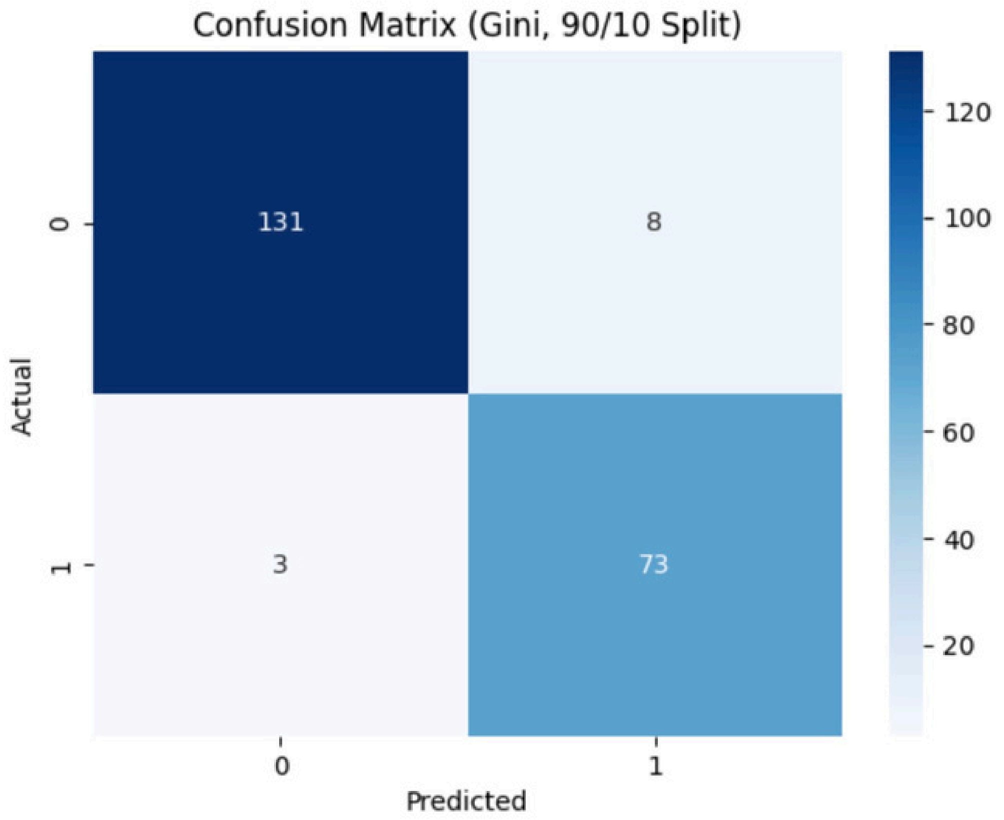
Gini index



accuracy: 0.9488



accuracy : 0.9465



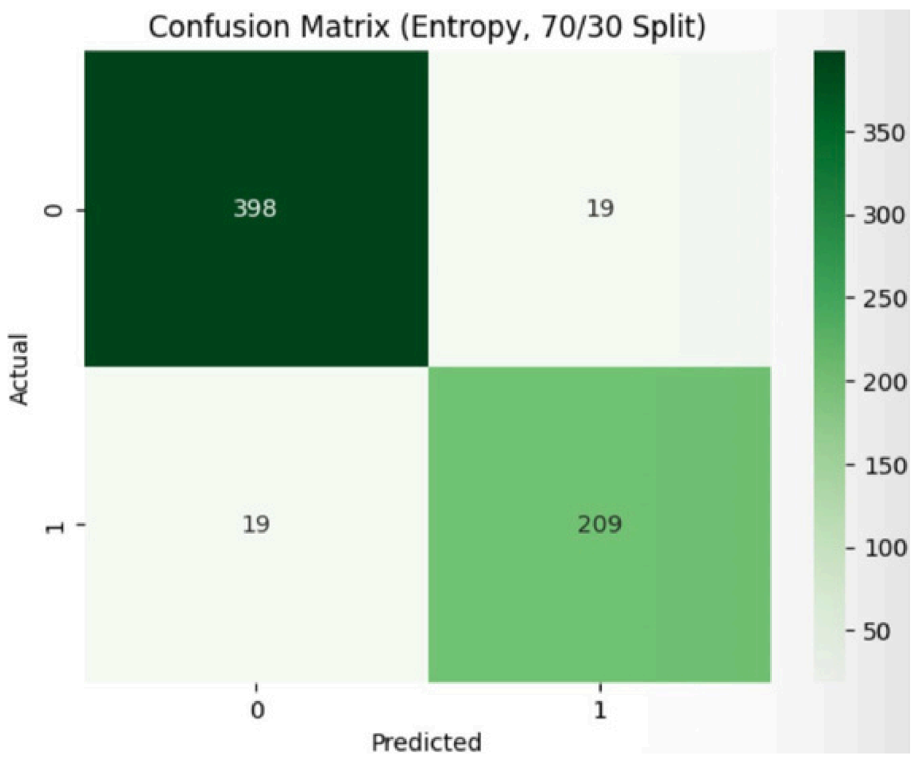
accuracy: 0.9488

The best Gini results were obtained each achieving an accuracy of 0.9488.

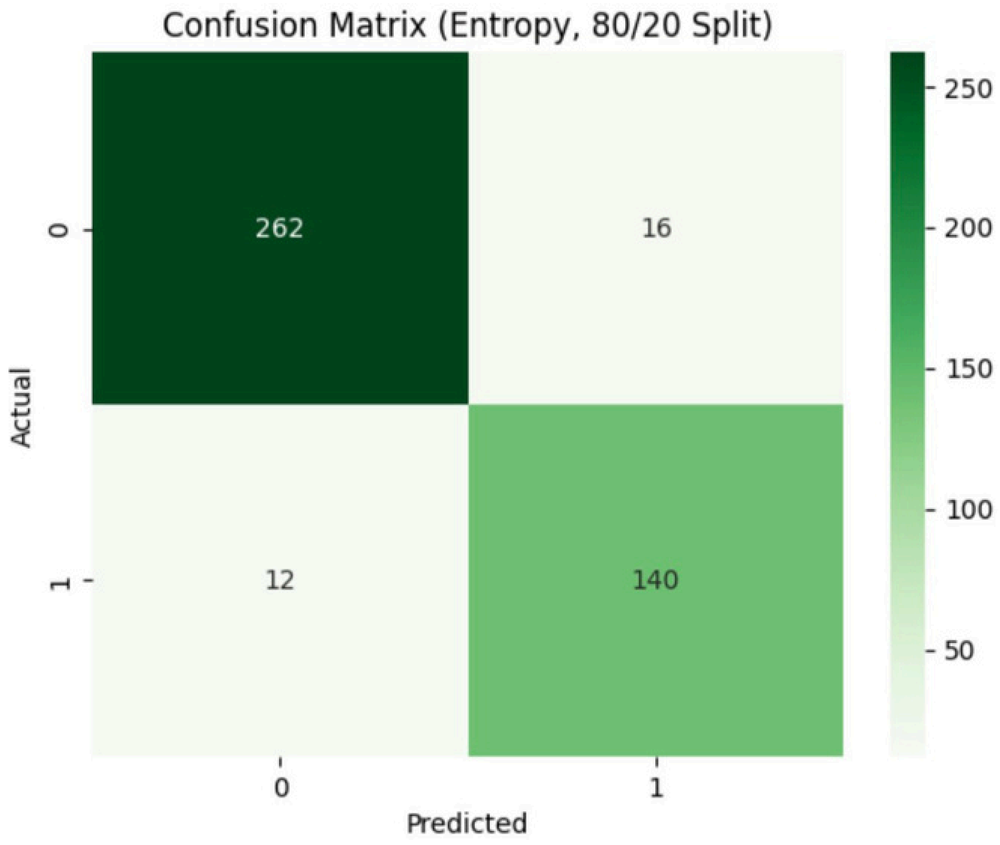
EVALUATION & COMPARISON

CLASSIFICATION RESULTS

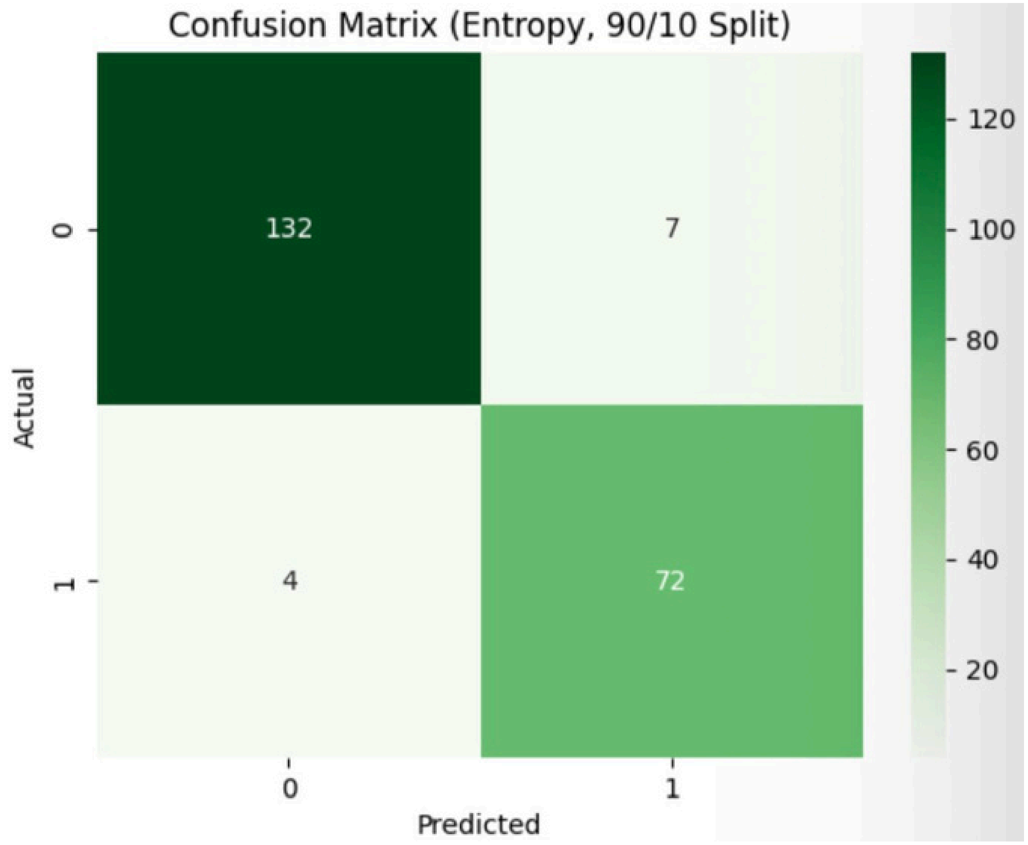
Information Gain (Entropy)



accuracy : 0.9411



accuracy : 0.9348



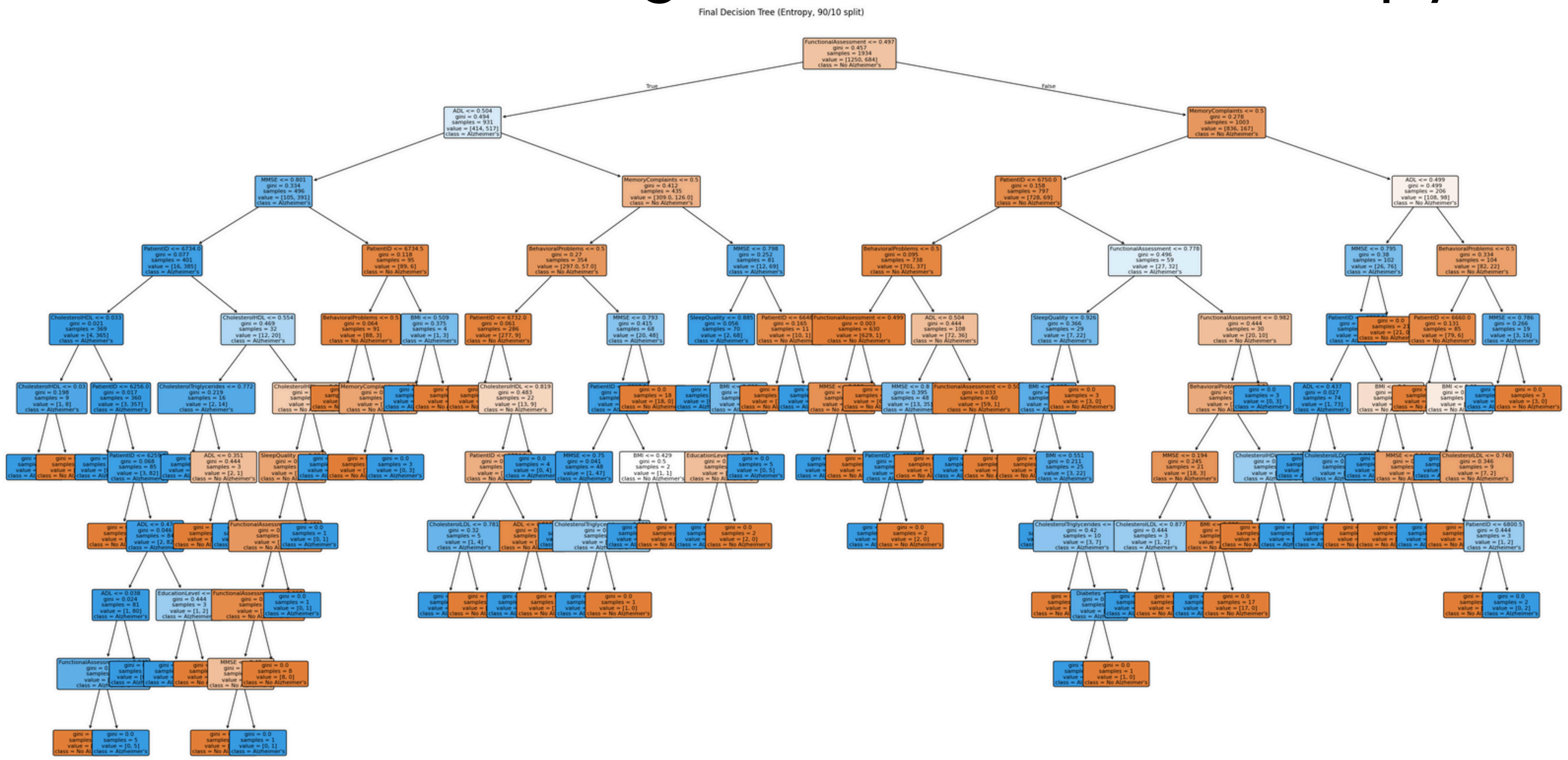
accuracy : 0.9488

The best Entropy configuration was the 90/10 split

EVALUATION & COMPARISON

CLASSIFICATION RESULTS

Best Overall Decision Tree Configuration (Gini vs. Entropy)



Overall, the best-performing and most stable configuration is Gini with the 90/10 split

EVALUATION & COMPARISON

CLUSTERING RESULTS

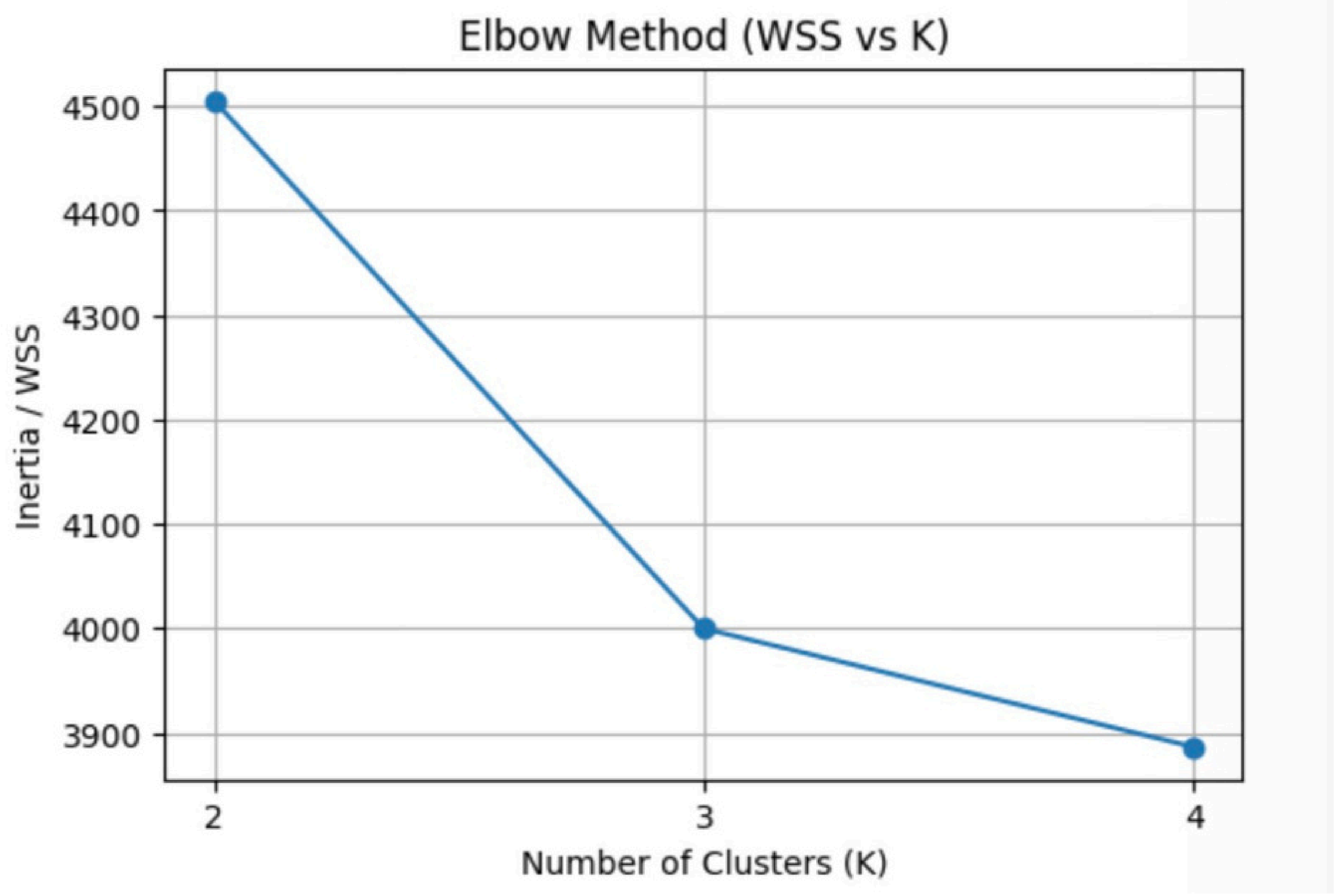
To determine the best value of K, we evaluated all three tested options (K = 2, 3, and 4) using three criteria:

- Inertia (WSS): Measures how compact the clusters are
- Silhouette Score: Measures how well-separated the clusters are

K	Inertia	Silhouette Score
2	4505.35	0.1076
3	3999.83	0.1155
4	3886.55	0.1051

EVALUATION & COMPARISON

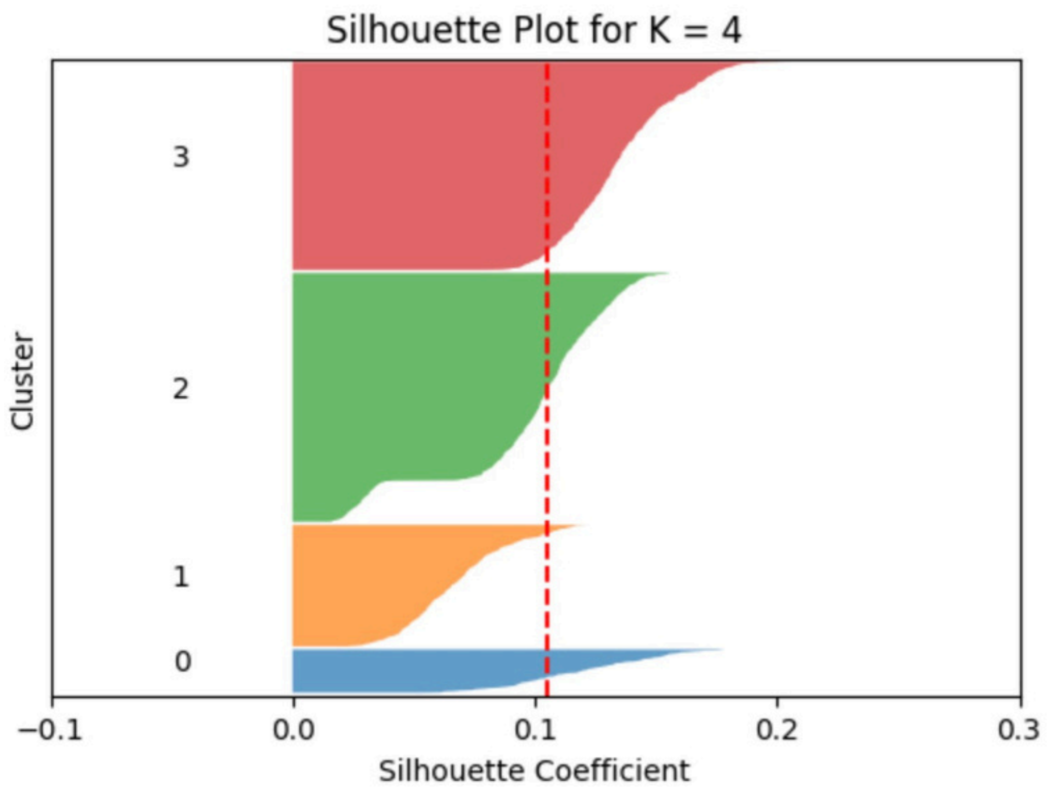
ELBOW METHOD



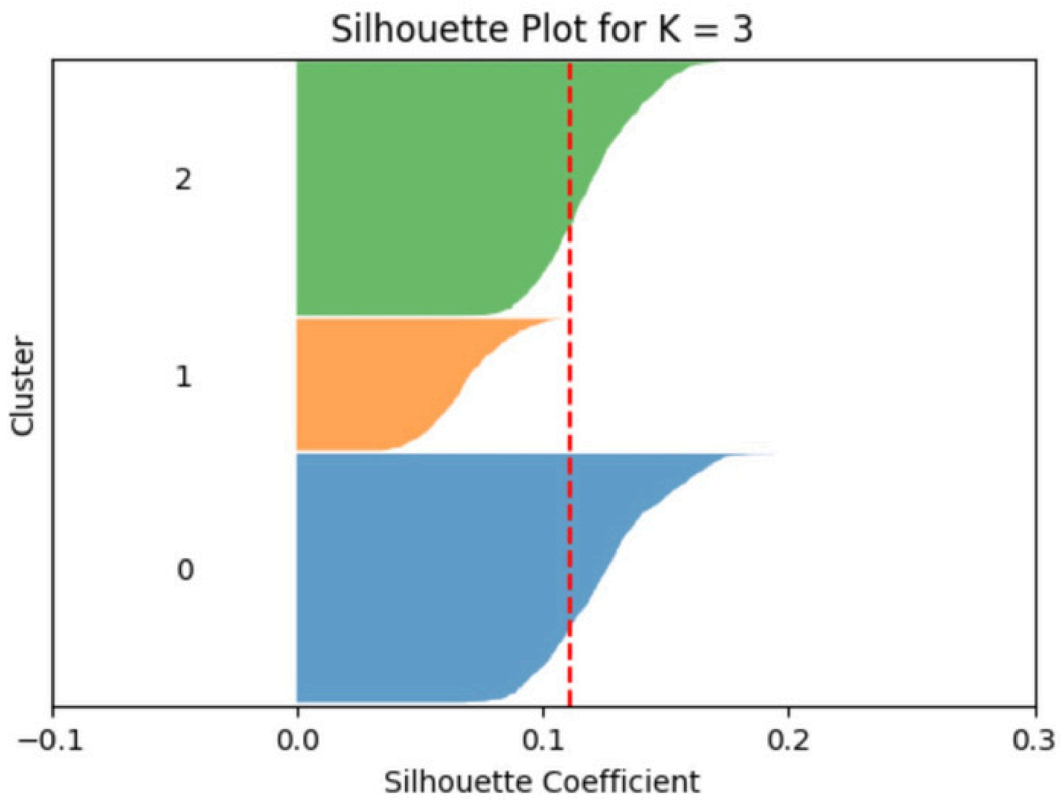
The Elbow plot shows that the decrease in WSS slows noticeably after K = 3

EVALUATION & COMPARISON

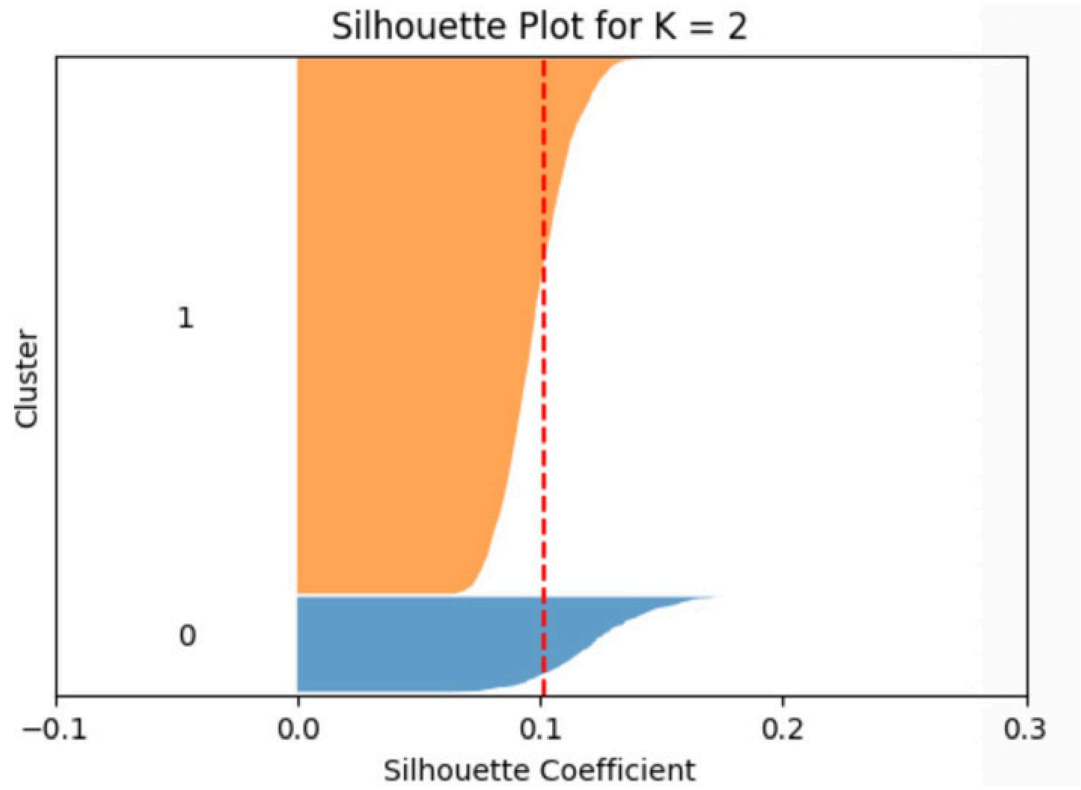
SILHOUETTE METHOD



Silhouette Score : 0.1051



Silhouette Score : 0.1155

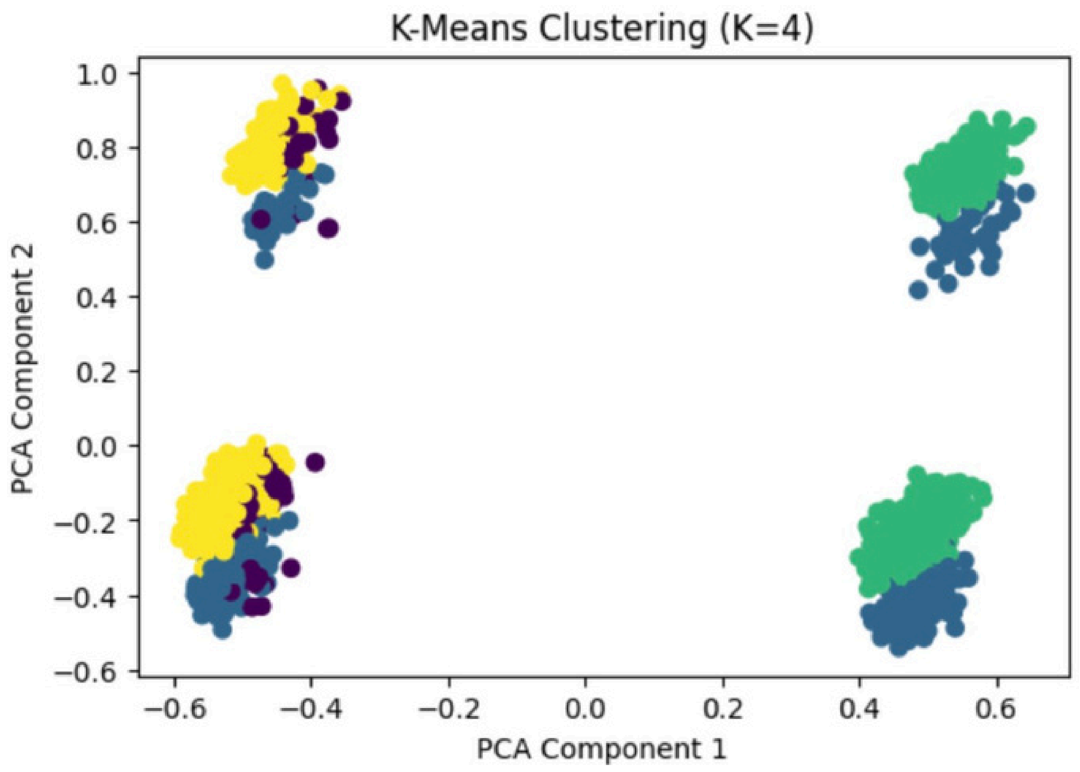


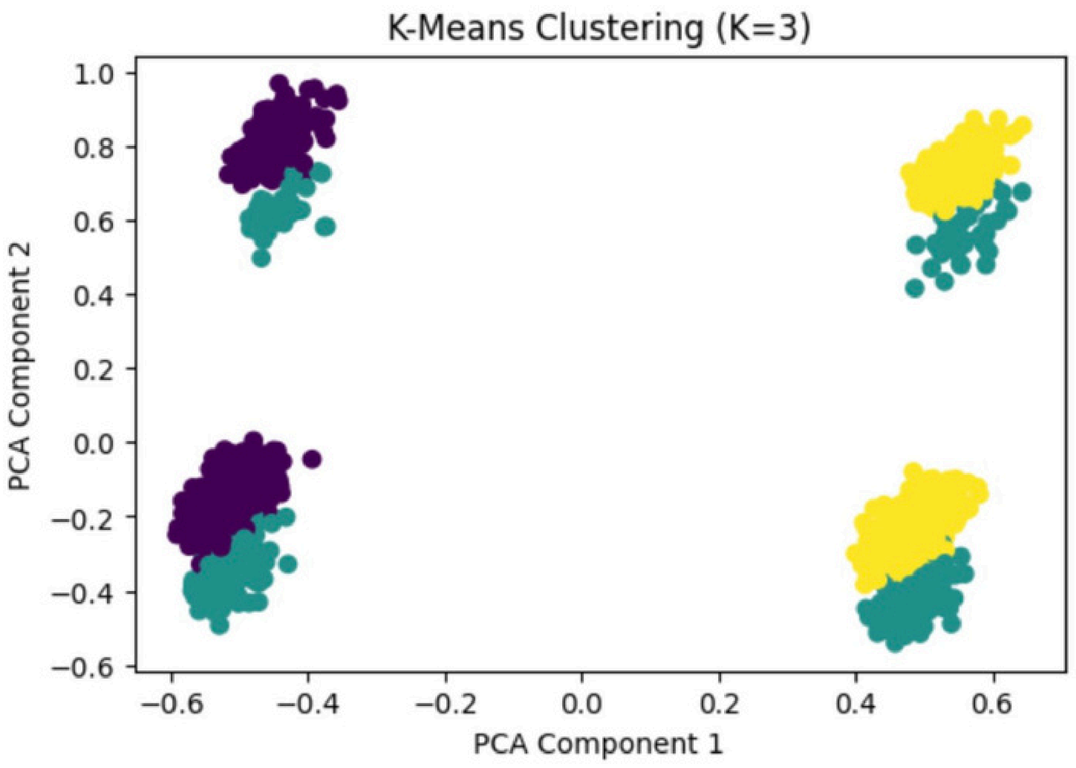
Silhouette Score : 0.1076

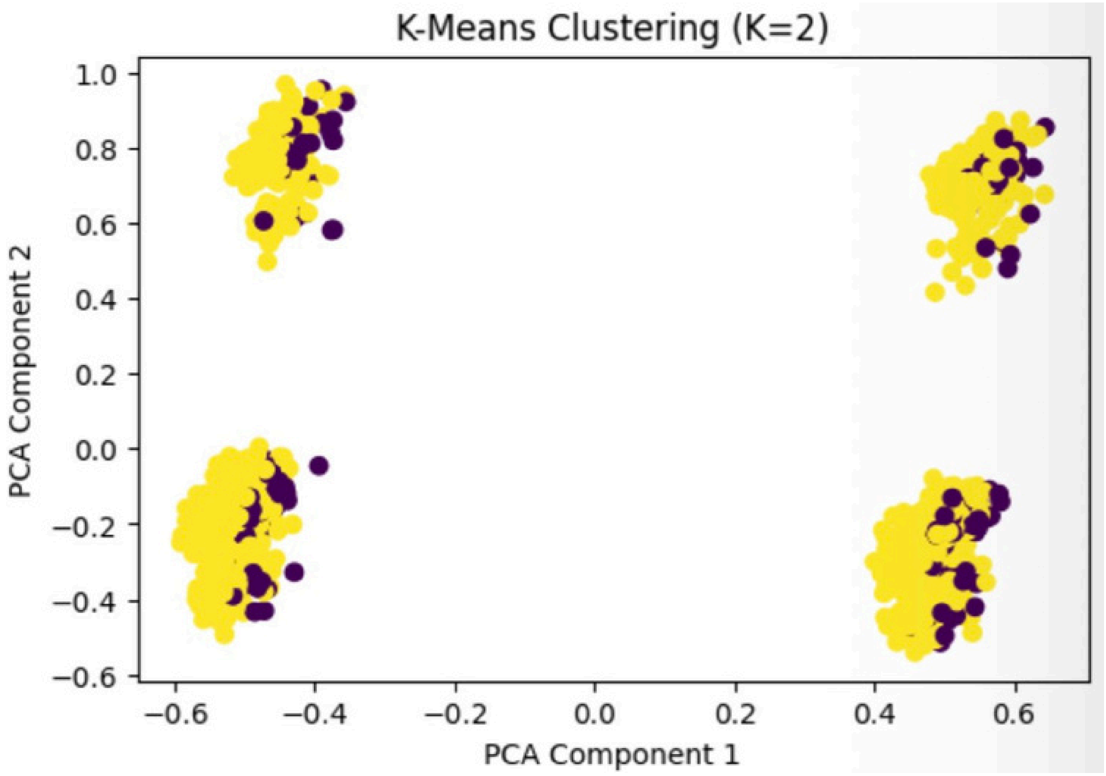
Silhouette Score (higher is better) — Best result: K = 3

EVALUATION & COMPARISON

K-MEANS CLUSTERING







Visual inspection of PCA plots — Best separation: K = 3

EVALUATION & COMPARISON

Selecting the Optimal Number of Clusters (Majority Rule)

The most effective clustering configuration is K = 3

	Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury	Hypertension	CholesterolLDL
0	3.108624e-15	0.433888	0.499826	0.504641	0.248219	0.128266	0.162708	0.093824	0.154394	0.492826
1	5.100671e-01	0.428784	0.522708	0.498670	0.228188	0.165548	0.138702	0.080537	0.134228	0.491668
2	1.000000e+00	0.424031	0.503859	0.516917	0.268605	0.148837	0.145349	0.097674	0.151163	0.498634

	CholesterolHDL	CholesterolTriglycerides	MMSE	FunctionalAssessment	MemoryComplaints	BehavioralProblems	ADL	Disorientation	PersonalityChanges
0	0.486946	0.514305	0.475806	0.499642	-2.109424e-15	0.156770	0.501095	0.152019	0.146081
1	0.510834	0.500644	0.496177	0.509472	1.000000e+00	0.149888	0.476655	0.167785	0.129754
2	0.490642	0.507871	0.505407	0.515824	-2.303713e-15	0.160465	0.506673	0.159302	0.166279

FINDING

A .CLASSIFICATION FINDINGS(Decision Tree)

The Decision Tree model demonstrated strong predictive performance and provided clear, interpretable decision rules, making it highly suitable for Alzheimer's prediction.

FINDING

B. Clustering Findings (K-Means)

K = 3 provides the most meaningful clustering structure and reflects natural differences in patients’ health and cognitive profiles.

FINDING

C. Best-Performing Models

Best Classification:
Decision Tree (Gini, 90/10) gave the highest accuracy and fewest errors.

Best Clustering:
K-Means with K=3 showed the best separation and highest Silhouette score.

FINDING

D. Relation to the Selected Research Paper

Our results match the paper's findings on key Alzheimer's risk factors and support using interpretable models like Decision Trees and meaningful clusters.

E. Overall Interpretation

Together, the Decision Tree and K-Means give a clear view of Alzheimer's patterns: accurate prediction and meaningful patient grouping. These results help support early detection and a better understanding of the disease.

Conclusion

We explored, cleaned, and analyzed the data, built a Decision Tree for prediction, and used K-Means to find meaningful patient groups. Overall, the project helped us apply the course techniques and understand Alzheimer’s patterns effectively.

- Data Understanding
- Data Preprocessing
- Classification (Decision Tree)
- Clustering K-Means)

References

[1] Y. Dubey, A. Bhongade, P. Palsodkar, and P. Fulzele, "Efficient Explainable Models for Alzheimer’s Disease Classification with Feature Selection and Data Balancing Approach Using Ensemble Learning," Diagnostics, vol. 14, no. 2770, pp. 1–15, Dec. 2024. doi: 10.3390/diagnostics14242770.

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[5] Our GitHub Project Repository (Phase 1–3), 2025.

Thank You

ANY QUESTIONS?